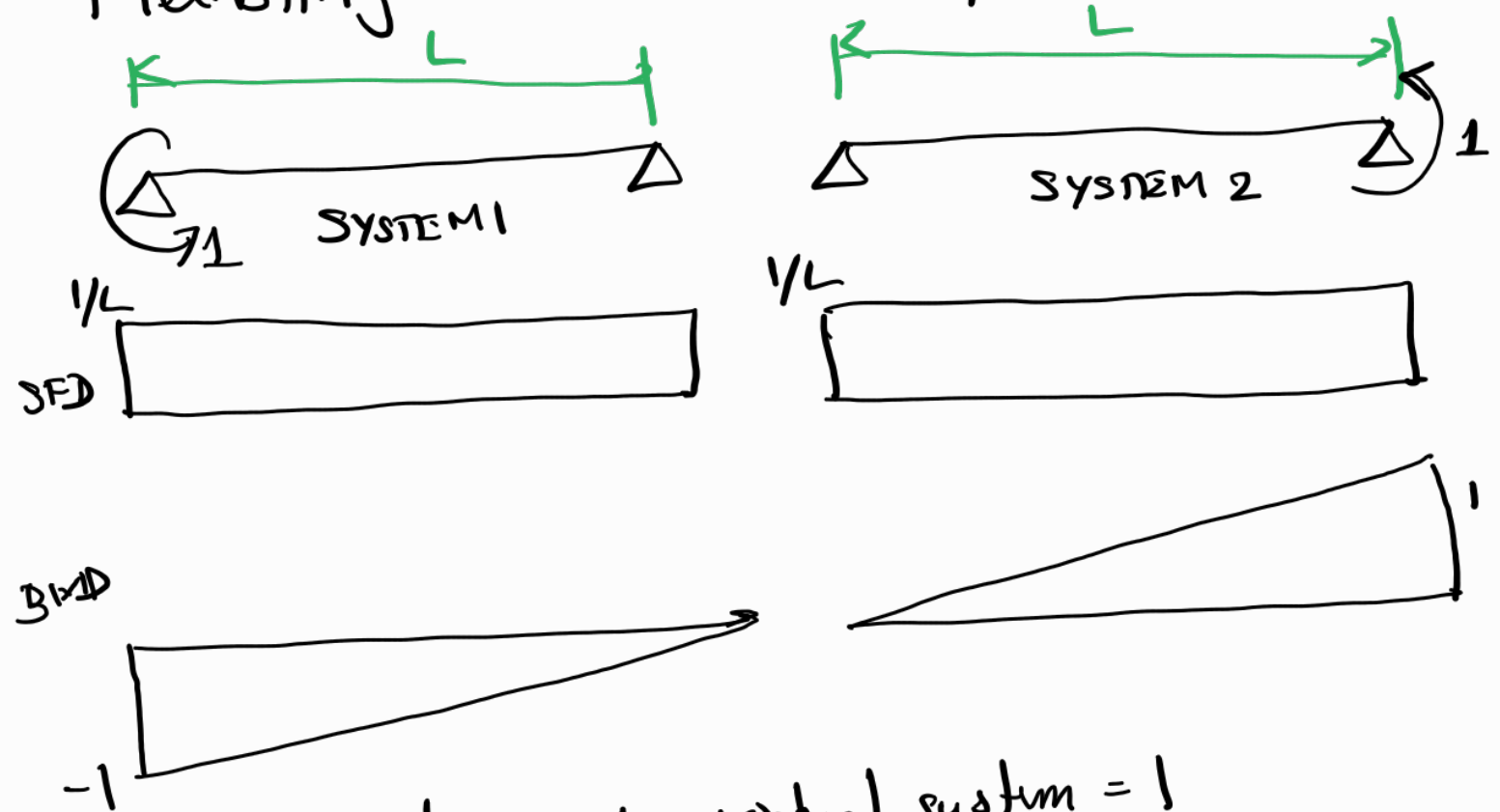


STEPS:

1. stiffness matrix in local system of simply supported beam
2. apply corotational transformation.



Flexibility matrix in XY plane.



a_{11} : real sys = 1, virtual system = 1

$$a_{11} = \int_0^L \frac{Mm}{EI} dx + \int_0^L \frac{Vv}{GA} dx$$

$$\begin{aligned}
&= \frac{1}{EI} \int_0^L \left(-1 + \frac{x}{L}\right)^2 dx + \frac{1}{GA} \int_0^L \frac{1}{L^2} dx \\
&= \frac{1}{3EI} \cdot L \cdot \left(-1 + \frac{x}{L}\right)^3 \Big|_0^L + \frac{1}{GAL} \\
&= \frac{L}{3EI} + \frac{1}{GAL}
\end{aligned}$$

$a_{21} \approx$ real sys = 1, virtual system = 2

$$\begin{aligned}
a_{21} &= \frac{1}{EI} \int_0^L \left(-1 + \frac{x}{L}\right) \left(\frac{x}{L}\right) dx + \frac{1}{GAL} \\
&= \frac{1}{EI} \left[-\frac{x^2}{2L} + \frac{x^3}{3L^2} \right]_0^L + \frac{1}{GAL} \\
&= -\frac{L}{6EI} + \frac{1}{GAL}
\end{aligned}$$

$$K(2:3, 2:3) = \begin{bmatrix} \frac{L}{3EI_{zz}} + \frac{1}{GA_y L} & -\frac{L}{6EI_{zz}} + \frac{1}{GAL} \\ -\frac{L}{6EI_{zz}} + \frac{1}{GAL} & \frac{L}{3EI_{zz}} + \frac{1}{GA_y L} \end{bmatrix}^{-1}$$

$$K(4:5, 4:5) = \begin{bmatrix} \frac{L}{3EI_{yy}} + \frac{1}{GA_z L} & -\frac{L}{6EI_{yy}} + \frac{1}{GA_z L} \\ -\frac{L}{6EI_{yy}} + \frac{1}{GA_z L} & \frac{L}{3EI_{yy}} + \frac{1}{GA_z L} \end{bmatrix}^{-1}$$

$$K(1,1) = \frac{1}{EA_x}$$

$$K(6,6) = \frac{1}{GJ}$$

T = Independent end forces $\rightarrow$ All end forces.

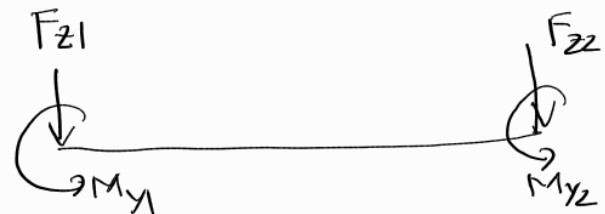
		N	M _{z1}	M _{z2}	M _{y1}	M _{y2}	T
F _{x1}	=	-1					
F _{y1}			1/L	1/L			
F _{z1}					-1/L	-1/L	
M _{x1}							-1
M _{y1}					1		
M _{z1}			1				
F _{x2}		1					
F _{y2}			-1/L	-1/L			
F _{z2}					1/L	1/L	
M _{x2}							1
M _{y2}						1	
M _{z2}				1			

Equilibrium:

$$M_{z1} + M_{z2} + F_{y2} \cdot L = 0$$



$$M_{y1} + M_{y2} - F_{z2} \cdot L = 0$$



Now transformation to global

Rotation matrix constructed using defn. of "orientation angle" used in LARSA.

- Member X axis lined up with joint1-joint2 line

$$\tilde{x} = \frac{\text{joint2} - \text{joint1}}{\|\text{joint2} - \text{joint1}\|}$$

- Member Z axis placed in plane containing joint1-joint2 line and global Z axis

$$-\tilde{y} = \frac{k \times \tilde{x}}{\|k \times \tilde{x}\|}$$

$$-\tilde{z} = \tilde{x} \times \tilde{y}$$

$$R_1 = [\tilde{x} | \tilde{y} | \tilde{z}]$$

- Rotate Θ = orientation angle about $\tilde{x}$.

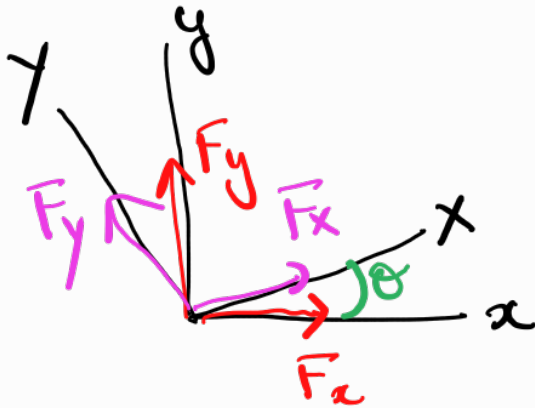
$$R_2 = \expm(\Theta \tilde{x}) \text{ — use Rodrigues for this}$$

- $R = R_1 R_2$

$$R = \text{blk diag}(R, R, R, R)$$

$$K_{\text{glob}} = R^T K T^T \boxed{R^T}$$

R^T or R ?



$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Equilibrium or static equivalence.

$$F_x = F_x \cos\theta - F_y \sin\theta$$

$$F_y = F_x \sin\theta + F_y \cos\theta$$

$$\begin{pmatrix} F_x \\ F_y \end{pmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{pmatrix} F_x \\ F_y \end{pmatrix}$$

This is R